

06.02 Polynomial Long Division

Common problems and solution methods

Rule guide: recognize the task first

Recognize	Apply these rules
Polynomial Structure And Equations	Look first for structure: degree, factor pattern, roots, or graph behavior.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

Example 1: Identify the model

Problem

$$x^2 + 7x + 12$$

How to solve it

Write the known quantities and restrictions.

$$a^2 - b^2 = (a - b)(a + b)$$

Check the result against the original statement.

Example 2: Apply the key rule

Problem

$$2x^2 - 5x - 3 = 0$$

How to solve it

Write the known quantities and restrictions.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Check the result against the original statement.

Example 3: Check the result

Problem

$$f(2) = 2^3 - 2(2) + 5$$

How to solve it

Write the known quantities and restrictions.

$$f(a) = 0 \Rightarrow (x - a) \text{ is a factor}$$

Check the result against the original statement.